

“Havin' money's not everything, not havin' it is”
The importance of financial satisfaction for life satisfaction in financially stressed households

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Abstract This paper explores the importance of financial satisfaction for overall life satisfaction in households whose principal source of subjective stress is financial. Using data drawn from 2 waves (2005, 2010) of the nationally representative General Social Survey in Canada, we find that for financially stressed Canadian households, their stress-affected sense of financial well-being overwhelmingly conditions their overall sense of life satisfaction. We consider as well the potential for financial stress to moderate the relationship between income and life satisfaction but find only a modest effect and then only for high income households in 2005.

Keywords: Subjective financial stress – Financial satisfaction – Life satisfaction – General Social Survey

JEL classification: D12 G02

1 Introduction and background

Kanye West expresses in the common vernacular what is well borne out by three decades of research into the relationship between absolute income and overall wellbeing – “havin’ money’s not everything.” As Easterlin (2015 4) summarizes, “for rich, developing, and transition countries, whether pooled or analyzed separately, there is no time series evidence that a higher economic growth rate increases the rate of improvement in life satisfaction.” Nor is there any evidence that “a greater increase in financial satisfaction accompanies more rapid economic growth,” (Easterlin 2015, 5). What receives less attention in scholarly research is the extent to which the sentiment expressed in West’s subordinate phrase: “not havin’ it is,” may also be supported by the data. Differentiating between life satisfaction and emotional well-being, Kahneman and Deaton (2010) find “less money is associated with emotional pain.” This paper contributes to the research on the financial determinants of subjective wellbeing (SWB) by examining the conditions under which financial satisfaction contributes to the overall assessment of one’s life satisfaction when financial concerns are present. A related consideration addresses an ambiguity in the literature about whether financial stress is itself a mediating or a moderating influence on the focal relationship between income and life satisfaction or if, instead, financial stress is simply an expression of financial “dissatisfaction.” Further, we examine these important issues in the context of a nationally representative data set. While the data constrain some of the questions we can ask, the results hold considerably greater external validity for those questions we can examine. Where much of the extant literature relies on idiosyncratic, dedicated surveys of a small group of participants to explore tentative relationships, our analysis of the data from a large nationally representative sample is a solid test of the strength of some of these tentative connections.

Well documented in the literature is the positive but only modest positive correlation between absolute income and social wellbeing, (Gasiiorowska 2014; Diener et al. 1993). These modest correlations are “difficult to explain” (Christoph (2010) and have motivated much of the research exploring alternative perspectives, such as the importance of relative income (Latif 2016, Brown and Gray 2016, Barrington-Leigh and Helliwell 2008), various additional factors and intermediate links. In addition to absolute income, Plagnol (2011) and Pereira and Coelho

(2013) examine the impact on financial satisfaction of assets and liabilities. Intermediate pathways either enhancing or eroding the effect of absolute income on overall well-being include perceived income adequacy, cognitive biases (pessimistically) distorting perceived income adequacy (Grable et al. 2013; Sahi 2017), financial stress (“money anxiety” Gasiorowska 2014) and inertia in subjective financial stress (Ayllón and Fusco 2017; Fusco 2016). With only a modest direct effect of absolute income on SWB, researchers have been exploring whether there are any indirect effects of income on SWB through the subjective assessment of one’s financial condition.

Understanding the relationship between material circumstances, perceptions of that circumstance and SWB connect the material and economic directly with the psychological – both cognitive and affective components. The substantial amount of research examining the effect of income on SWB has found that “material factors are strongly related to the cognitive component of SWB” (Ng and Deiner 2014, 326; Diener et. al. 2010). Both the psychology and family counselling research have looked closely as well at the affective components of “money attitudes.” The “common finding is that there are roughly orthogonal factors relating to (a) the affective aspects of money, such as distrust, anxiety, power, prestige, esteem or achievement, and relating to (b) economic aspects such as budgeting, planning, spending retention, or debts,” (Gasiorowska 2014, 66).

Researchers initially decomposed SWB into three distinct elements: life satisfaction, pleasant and unpleasant affect (Diener 1984). Diener et al. (1999) later added domain satisfaction, such as satisfaction with one’s finances, one’s job or one’s health, as principal, rather than derivative, components of SWB. Some of the debate about the key determinants of SWB centers now on whether life satisfaction, as a “cognitive, global evaluation of one’s life” (Ng and Deiner 2014) is an aggregate of independently determined levels of satisfaction in the domains of work, health and finances or if, instead, domain satisfactions are independent determinants of SWB, together with but separate from life satisfaction. Plagnol (2011, emphasis added) financial satisfaction is “an integral *component* of overall life satisfaction and well-being.” Johnson and Krueger, (2006, 680), suggest too that it is “satisfaction with one’s situation, rather than the financial situation itself, that is associated with life satisfaction...perceived financial situation is the environmental variable that mediates between income and life satisfaction.” Some suggest instead that domain satisfaction is an important determinant of SWB, separate from life satisfaction, (van Praag, Frijters and Ferrer-i-Carbonell 2003; Heller, Watson and Ilies 2004; Easterlin 2006). Ng and Deiner, (2014, 326) argue, however, for a more nuanced explanation through multiple pathways. “[E]ffects of SWB correlates may act via different processes, resulting in their differential relations to different SWB facets.” Despite considerable effort, then, scattered as it is across the literature in economics, psychology, consumer finance, and family counselling, there remains an incomplete understanding of how perceptions of one’s financial situation may affect one’s sense of financial well-being and overall sense of life satisfaction. Since with our data we can only examine the determinants of life satisfaction and not the broader SWB concept, we inquire into the manner in and extent to which the subjective assessment of one’s financial situation affects one’s life satisfaction.

1.1 Financial satisfaction, financial stress and other indicators of the subjective financial condition

The contested relationship between financial satisfaction, financial stress and other indicators of the subjective financial condition mirrors the above debate about the location of financial satisfaction in the pathway relationship between income and SWB but it is complicated further by an ambiguity in the terminology. Across the literature, the financial condition is variously represented by “financial satisfaction,” “financial stress,” and “perceived income adequacy (PIA).”¹ For example, where the European Social Survey asks respondents how they feel about current income and whether it is perceived to be adequate (as implicit in the answer options), the Canadian Social Survey asks about financial stress and financial satisfaction. How then are “perceived income adequacy” “financial stress” and “financial satisfaction” related? As Prawitz et al. (2006, 34) note: “there has been little agreement as to the best way to measure [subjective assessment of the financial situation], or even which construct was being measured.” It remains the case more than a decade later “that no instrument exists that measures the construct encompassing

¹ Confounding the disputed conceptual relationship is the considerable ambiguity in the meanings of the constructs being measured. This may be due in part to the challenge presented by the multi-disciplinary nature of this research where researchers in one discipline adopt vocabulary that bumps up against discipline-specific conventions in another. For example, where economists would make a clear and consistent distinction between income (as revenue per period of time that excludes debt) and wealth (as the stock of assets net of liabilities which can generate income but is valued independently of time and accounts for debt), psychologists may not.

perceptions about financial well-being and stress about one's financial condition and that has undergone a rigorous process to test for content, construct and criterion validity, as well as for reliability," (*op cit*, 36).

In some studies, financial stress and PIA are separate concepts and play and distinct roles in determining financial satisfaction. If "strain" and "stress" may be considered synonymous, Sears, as quoted in Sinclair et al. (2010, 3) identify financial strain (stress) as an affective construct "consisting on negative emotional reactions related to one's current financial status" and PIA, in contrast, is "a cognitive construct consisting of evaluations of one's current and expected future ability to afford basic living needs (PIA-needs), and the ability to afford current and future lifestyle desires (PIA-wants)."

Much of the research has focused on financial stress as a mediating influence on financial satisfaction, (Gasiorowska 2014; Joo and Grable 2004; Bailey et al. 1998). For others, financial stress is a critical determinant of financial satisfaction with financial attitudes [PIA] potentially mediating between the financial stressors (e.g., income) and subjective financial stress (Joo and Grable 2004; Porter and Garman 1993). Pereira and Coelho (2013, 1000) bypass financial stress and instead focus on the critical mediating effects of perceived income adequacy and borrowing constraints on the relationship between absolute income and life satisfaction directly. Prawitz et al. (2006, 36) contrariwise, place financial stress on a continuum with financial satisfaction suggesting these are opposite and mutually eclipsing concepts on "a continuum extending from negative to positive feelings about and reactions to the financial condition." Similarly, Porter and Garman (1993, 159) appear to treat financial stress as the absence of financial satisfaction when they talk about one's desire to "avert financial difficulties and increase perceived financial well-being." In this paper, we explore the extent to which financial stress mediates the relationship between income and life satisfaction. In view of the broader terminological ambiguity, we consider whether financial stress is a determinant of financial satisfaction or if financial stress is equivalent to financial dissatisfaction as the opposite of financial satisfaction.

The evidence of an indirect income effect through PIA or financial stress on life or financial satisfaction (or first on financial satisfaction then out to life satisfaction) is mixed, very possibly due to the issues with construct validity discussed above. Pereira and Coelho (2013, 1000) find that the effect of perceived income adequacy on life satisfaction may be nearly identical for low and high income households, for those reporting an ability to at least cope on current income. Contrary to the prior that individuals with a fragile financial situation would be more responsive to perceived income (in)adequacy, they find, in fact, limited evidence to support the suggestion that lower (higher) absolute income translates into lower (higher) perceived income adequacy. Gasiorowska (2014, 71) on the other hand finds strong evidence in support that higher absolute income "led to significantly lower money anxiety...and lower anxiety significantly increased subjective wealth," where "subjective wealth" (using wealth as she does interchangeably with income) is measured as an index of responses to 3 questions related to how participants feel about their financial situation; their ability to make ends meet; and whether income adequately fulfilled needs and wants. As framed by the available Canadian data, our question is does absolute income affect subjective financial stress, where such stress may be an indicator of perceived income inadequacy?

1.2 Financial stress as a moderating influence between income and life satisfaction

The potential for financial stress to act as a moderating influence of financial stress in respondents' assessments of financial satisfaction has received less attention. Recent work in both behavioral finance and statistical analyses of survey data suggest the possibility of a moderating influence of the financial condition on levels of satisfaction. A mediating variable (M) is variable that explains how X affects Y because X affects M and M affects Y. A moderating or conditioning variable (W) is a variable that affects the strength of the relationship between X and Y, conditional on the value of W.

The potential for a moderating or conditioning effect of stress on perceptions of life satisfaction extends recent work in behavioral finance on the effect of "tunneling" (as the narrowing of one's attention, Mullainathan and Shafir 2013) and is evident in the way survey design can focus a respondent's attention on stress ...influences respondents' assessment of overall life satisfaction when specific domain question precede questions about one's more general life satisfaction, (Schwarz, Strack and Mai 1991). Bonikowska et al. (2014, 626) in their analysis of the Canadian GSS suggest that negatively-oriented questions on a "time-crunch" heightened respondents' attention on this stressor and conclude that such "questions preceding the question on life satisfaction in the 2005 and 2010 GSS may account for the lower level of life satisfaction observed in those years." In this paper, we consider whether there is any moderating effect of financial stress on the relationship between absolute income and financial satisfaction as a domain of life satisfaction.

1.3 Importance

Where financial policy makers tend to focus on objective assessments of financial stress such as debt service to income ratios and other measures of financial fragility, the importance of subjective financial satisfaction and subjective indicators of financial stress receive much less attention by them. Yet, Keese (2012) for example, explores differential perceptions of debt for a given objective debt burden and finds subjective debt burdens contain more information than is available by looking at only an individual's current economic situation. Several studies suggest, further, that subjective assessments of financial stress influences objective financial behavior in a way that can materially alter the objective fragility of a household's finances. Perceptions of one's financial situation affect one's financial behavior, (Danes and Rettig 1993). Differences in money attitudes may have, as Hayhoe et al. (2012) demonstrate, important implications for understanding savings and consumption behavior. Garðarsdóttir and Dittmar (2012) demonstrate the connection between materialism, overspending and consumer debt levels in the case of Iceland.

Several studies in psychology and behavioural finance now underscore the importance of disentangling these potential influences for understanding both the potential threats of household debt to our overall economic and our overall psychological sense of well-being as well as the interaction between the economic and psychological for designing effective interventions. Educating to promote financial literacy and financial capability is now one of the primary interventions promoted by several national and international organizations, where research suggests there is a positive impact of financial literacy education on financial behavior, financial stress and financial satisfaction (Lusardi 2008; O'Neill et al 2005; Hira and Loibl 2005; Hilgert, Hogarth, and Beverly 2003). Notably, however, the evidence remains mixed; while enhanced financial literacy is thought to increase financial control, financial control may not, in fact, reduce financial stress (Gasiorowska 2014) and so the question of the extent to which increasing absolute income reduces financial stress and increases overall life satisfaction is still an open one.

If financial stress is a strong mediating variable, increasing absolute income will reduce financial stress and increase overall life satisfaction. If this relationship is relatively constant over household income levels, reducing financial stress and increasing life satisfaction will be achieved by policies that promote balanced economic growth. If instead the mediating role of financial stress on the relationship between income and life satisfaction varies across income levels, policies that target income redistribution may be more effective than growth policies for increasing overall life satisfaction.

Finally, if financial stress is a moderating variable conditioning negatively the relationship between absolute income and life satisfaction (i.e., the more financial stressed, the lower the effect of absolute income on life satisfaction), the intervention should directly target the reduction of financial stress by means other than increasing or redistributing absolute income.

The relationship between financial stress and financial satisfaction, between financial satisfaction and income, and finally between financial stress and overall life satisfaction remain unclear. The aim then of our analysis is to shed light on the following questions. Is one's financial situation an important determinant of one's overall life satisfaction? Does absolute income affect financial stress? How are financial stress and financial satisfaction related? Is financial stress best understood as (a) a mediating variable between income and satisfaction (financial and life), and/or (b) a moderating variable between income and satisfaction (financial and life), and/or (c) a definitional alternative to financial satisfaction (i.e., synonymous with "financial dissatisfaction")?

2. Data and Method

2.1 Dataset description

We use data from 2005 and 2010 waves of the nationally representative General Social Survey (GSS).

Statistics Canada's General Social Survey (GSS) has two primary objectives: (i) to gather data for the purpose of monitoring changes in the living conditions and well-being of Canadians over time; and (ii) to provide information on specific social policy issues of current or emerging interest. ...The target population

of the GSS is non-institutionalized persons 15 years of age or older residing in Canada's ten provinces. Data are collected by means of computer-assisted telephone interviews. Households are selected through Random Digit Dialing methods, and one household member (aged 15 or older) is randomly selected to participate in the survey. Interviews by proxy are not allowed. (Statistics Canada 2013)

Our primary variables of interest are "financial stress," "financial satisfaction" and "life satisfaction." Financial stress appears in the GSS as a response to the question "*What is your main source of stress?*" with "*financial concerns*" one of 5 possible answers. Both financial satisfaction and life satisfaction are explored as single questions using a ten-point dissatisfaction-satisfaction scale. ("*Please rate your feelings about...your finances.*" "...*how do you feel about your life as a whole right now?*")

For all those respondents who identify high stress levels (3 or higher on a 5 point scale), we first distinguish between those respondents who rank stress from financial concerns as their primary source of stress and those who identify another (job, health or other) primary stressor. Notably, those for whom financial stress is not the *primary* source of stress may nevertheless be feeling financially stressed but will be captured here in the "non-financial stress" group.

There are 19,559 observations in 2005 and 15,227 in 2010. In order to account for the unobservable time and province specific effects we cluster standard errors in all of our estimation on province-year level. In 2005 and 2010 the respondents were also asked to rank their overall satisfaction with life on a scale of one to ten. Less than 3.5% of respondents failed to provide this information. Additionally in 2005 but not in 2010 the respondents were asked to rank their financial satisfaction on the same scale.

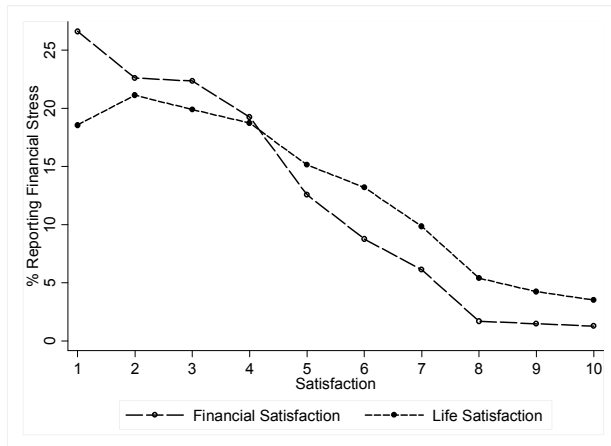
In general, we cannot with these data address additional considerations that may be relevant. Neither wealth nor any disaggregation of assets and liabilities are probed in the GSS. Personality characteristics such as overall pessimism-optimism moderating one's assessment of stress and overall life satisfaction are similarly excluded (Ng and Deiner 2014). Relative income (D'Ambrosio and Frick 2007; Dolan, Peasgood and White 2008) and per capita adjustments (Hsieh 2004) are not possible to analyse with these data, though notably household composition is reflected in household size, presence of children, and whether it is a multigenerational household.

2.2. The Stress-Satisfaction Connection

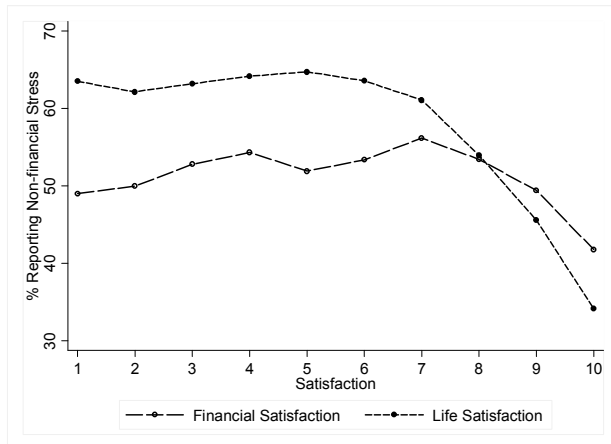
The association between financial stress and satisfaction can be seen in the following graphs. For the years 2005 and 2010 financial satisfaction closely tracks life satisfaction and both appear to be strongly negatively associated with financial stress in households reporting high financial stress as their primary source of stress (Graph 1). The association appears as well in those reporting either primary stress from other, non-financial sources (Graph 2) or no high levels of stress (Graph 3) but by visual inspection at least, the association appears to be looser. This preliminary observation invites closer examination.²

² Notably, respondents reporting high non-financial stress as their primary source of stress may be experiencing financial stress as well.

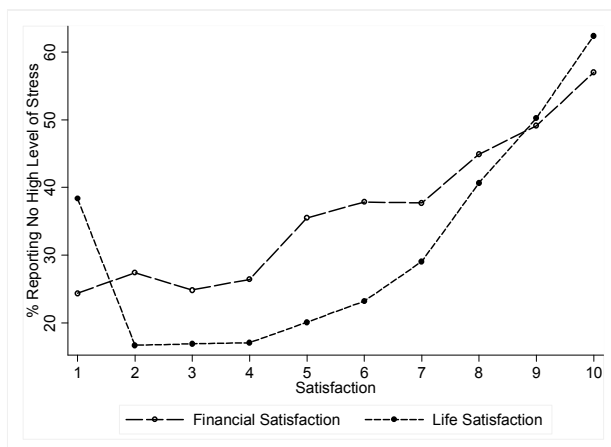
Graph 1. Financial Satisfaction and Life Satisfaction for high financially stressed households



Graph 2. Financial Satisfaction and Life Satisfaction for other highly stressed households



Graph 3. Financial Satisfaction and Life Satisfaction for low/no stress households



2.3 Model estimation

We start from the premise that one's subjective financial situation is influenced in part by one's income and in part by other sources of financial stress (such as debt levels) unobservable in the data examined here. We seek to understand better how subjective assessments of one's financial situation as measured by subjective financial stress and financial satisfaction relate to assessments of one's overall satisfaction. Specifically, our primary questions are if and how (i.e., mediation) and possibly when (as moderation) high financial stress affects life satisfaction. A second and related question is whether financial stress is an expression of or distinct from financial dissatisfaction. To explore these questions, we adapt Hayes' (2012) regression estimation process of a simple mediation/moderation model to a probit estimation, as applicable to our data.

There are 4 dimensions of our analysis:

- (1) Income affects Stress (Financial and Non-financial)
- (2) Income affects Satisfaction (Financial and Life)
- (3) Income and Stress affect Satisfaction
- (4) Income and Stress affect Satisfaction both directly and through a moderating effect

To investigate the effect various covariates have on stress we estimate multinomial probits with financial stress and non-financial stress as dependent variables. Embedded in this analysis of the role of Stress is the assumption that Stress is linearly related to Income, with an error (ϵ). We test the following relationship using a multinomial probit estimation of Specification (1). Results are reported in Table 1 below.

$$\text{Stress} = \text{Constant} + \alpha_1 * \text{Income} + \gamma_1 * \text{Other} + \epsilon_1 \quad (1)$$

"Other" captures the vector of other covariates (income as well as other socio-economic and demographic variables) considered in the explanation of Satisfaction (Financial or Life). We use the following covariates: dummies for high and low education, dummies for age of respondent being under 25 and above 64 years, being an income provider (either sole income provider or one of multiple providers), labour force status, union membership, sex, marital status, presence of children in the household, dummy for multigenerational household, household size, a full set of provincial and time dummies, urban residence, dummies for primary source of income, dummies for low and high household income³ and finally a dummy for interviews conducted on the weekend. For parsimony in reporting, we suppress some of these results in the tables below.

The effects of income and stress on life and financial satisfaction are similarly modeled. Our model captures the total effects of Income (Specification (2)) and the direct effect of income (α_2) allowing for a separate indirect effect (β_3) through Financial Stress (Specification (3)) on the dependent variable Satisfaction in two linear models:

$$\text{Satisfaction} = \alpha_2 * \text{Income} + \gamma_2 * \text{Other} + \epsilon_2 \quad (2)$$

$$\text{Satisfaction} = \alpha_3 * \text{Income} + \beta_3 * \text{Stress} + \gamma_3 * \text{Other} + \epsilon_3 \quad (3)$$

where Satisfaction is modeled for both Financial and Life Satisfaction in the results reported in Table 2 below; stress is estimated as financial and non-financial stress. We are able to use both waves (2005 and 2010) of the GSS to examine the effects of the co-variables on Life Satisfaction; the analysis of Financial Satisfaction is constrained, however, by the available to data to only the 2005 cycle. To analyze these satisfaction variables we estimate two sets of ordered probits. First, we use all the covariates we used in stress probits; second, we add directly the sources of stress into model specification (3) as indicated above.

³ The income information included in the surveys is rudimentary. We have access to nominal household income brackets, only. This presents an obvious problem. Since we are not working with a continuous variable we cannot use a price index to deflate nominal values into constant dollars. Instead for each survey year we create dummy variables that indicate if a household nominal income falls below twenty thousand dollars or above one hundred thousand dollars.

To test the significance of relationship described by (4), we need to modify Specification (3) and do so in the manner suggested by Hayes (2012). Where Financial Stress itself may be affected by Income, there is the possibility of an interaction between Income and Financial Stress. Where the level of income affects the strength of the effect of Financial Stress on Satisfaction, i.e., there is a differential effect of Financial Stress dependent or conditional on the level of income, Financial Stress is moderating income's effect on Satisfaction. In this case, if Financial Stress is a moderator of income, we would expect to see a statistical significance in the coefficient (δ) in the following Specification, which is equivalent to Equation (1) in Hayes (2012):

$$\text{Satisfaction} = \alpha_4 * \text{Income} + \beta_4 * \text{Stress} + \gamma_4 * \text{Other} + \delta_4 * (\text{Income} * \text{Financial Stress}) + \epsilon_4 \quad (4)$$

Since Hayes employs regression analysis, he is able to measure the cardinal value for the indirect effect of (for our model) the product of Income*Financial Stress such that he can draw conclusions about how much a one unit change in Income will multiplicatively in total affect Life Satisfaction. The combined use of multinomial probit (1) and ordered probit (3) analysis of our study precludes such precision. Where Hayes is able to verify the direct (α_3), indirect ($\alpha_1 * \beta_3$) and total effects (α_2) independently, we are able to verify only the direct (α_3) and total effects (α_2) of income on Life Satisfaction.

3. Results and discussion

3.1 Income affects Life Satisfaction only modestly

The coefficients on the income variables in Specification (3-L) in Table 2 for Life Satisfaction reveal a consistent but only modest direct effect of Income on Life Satisfaction. In 2005, the probability that higher income corresponds to higher life satisfaction (given the values of the default categories) increases from (-0.205) to (0.101); in 2010 from (-0.116) to (0.153). In both years higher income corresponds to higher life satisfaction and lower income corresponds to lower life satisfaction. The low income coefficients are (-0.205) in 2005 and (-0.116) in 2010. The high income coefficients are (0.101) and (0.153) respectively, with all statistically significant. The total effect of income on Life Satisfaction derived from Specification (2-L) is greater in absolute terms than derived from Specification (3-L) in 3 of the 4 cases suggesting that there is an indirect effect on life satisfaction of income as well in all but the 2010 high income case.

3.2 Income affects financial stress only modestly and inconsistently.

The first column of Table 1 presents results for respondents who report high levels of stress (3 or higher on a 5 point scale) and who identify financial considerations as a primary source of stress (1-FS). For these respondents, income has a statistically significant strong effect for only higher income households: 2005 (-0.330), 2010 (-0.406). For households with a nominal income of less than \$20K, absolute income on financial stress is statistically insignificant.

These results suggest that high income modestly lowers the probability of reporting financial stress as the primary stressor. For low income earners whose primary source of stress is financial, the probability of reporting financial stress is unaffected by the household's low income status. In other words, having a low income does *not* affect the probability that low income earners are reporting financially stress.

3.3 Financial stress has limited mediating effect on the income – life satisfaction relationship

The results reported above in 3.2 suggest financial stress can have only a modest and inconsistent mediating effect. Comparing the coefficients on the income co-variables in Specification (2-L, without financial stress) with Specification (3-L, with financial stress) further supports the conclusion that financial stress has a minimal mediating effect between income and life satisfaction. The addition of the financial stress co-variate in Specification (3-L) alters the coefficients on the income covariates only minimally, and then in only 3 of the 4 cases, with the 2010 high income coefficient remaining constant at (0.153). All coefficients in these specifications are statistically significant.

3.4 Direct effects of financial stress on life satisfaction are considerable and considerably more than non-financial stress.

While financial stress may not be mediating consistently between income and life satisfaction, the mere presence of financial stress reduces life satisfaction considerably. In Specification (3-L), respondents experiencing high financial

stress are almost certainly (0.864) less likely to report high life satisfaction. This effect is substantially greater than employment, education or marital status.

In all specifications where both kinds of stress are included, coefficients for financial stress have greater absolute magnitudes than coefficients for non-financial stress. Again keeping in mind the construction of our variables, we observe in these data that financial stress has more severe negative effects on life satisfaction than non-financial stress. Recalling our construction of the variables though we note that the non-financial stress cannot be literally or simply interpreted; rather, those reporting other “non-financial” stress as the *primary* source of stress does not preclude the possibility that financial stress is affecting life satisfaction for these respondents.

Notably adding financial stress directly decreases the coefficients on age, employment status and number of young children in the household, increases the effect of sex, and leaves unaffected marital status. Adding non-financial stress has only a modest negative effect on financial satisfaction (-0.256, from Specification 3-F) but its negative effect on life satisfaction more than doubles in absolute terms (-0.563).

3.5 Financial stress and financial dissatisfaction are tightly linked

In Specification (3-F), the coefficient on financial stress equals (-1.030). Respondents were, with certainty reporting lower financial satisfaction when financial stressed. This result suggests that while financial stress and financial dissatisfaction may not be synonymous, they are very tightly linked.

The introduction of financial stress in Specification (3-F) exposes the direct effect of income on financial satisfaction, which remains moderate (-0.431) for low income and (0.239) for high income households. These coefficients are significant statistically and more than twice the effect of income on life satisfaction for 2005, the only year we have to analyse financial satisfaction.

Notably, the combined results reported in (3.1) above and these results present something of a puzzle. As reported in Table 1, income modestly affects financial stress but only for high income households. Where the results here suggest financial stress is tightly linked to financial dissatisfaction, we would expect the income results to carry through. Instead, we see the modest probability of reporting financial satisfaction statistically significant for both high *and* low income households.

3.6 Minimal moderating influence of financial stress on the link between absolute income and overall life satisfaction

The coefficients on the interaction terms (Stress*Income) to capture the potential moderating influence of financial stress in Specification 4-L suggests financial stress has little moderating influence on the relationship between income and either life or financial satisfaction. The interaction term of financial stress and income is statistically significant for high income households in 2005 only. The interaction term of non-financial stress and income is statistically significant for high income households in 2005 and low income households in 2010. Such mixed results preclude any firm conclusions. Notably though the presence of the interaction term alters both the size and the significance of the income variables directly in both Specifications but yields little change on all non-income coefficients.

3.7 Other observations

Consistent with much of the existing literature, we find Canadians exhibit many of the same characteristics as respondents in other countries. Age correlation is consistent with life cycle results where youth (ages 15-24) and seniors (55 years of age and older) perceive less financial stress, marital status correlates with subjective financial stress where married respondents perceive less financial stress, and the presence of children under the age of 19 correlates positively with subjective financial stress.

4. Conclusion

This study explored the relationship between financial stress, financial satisfaction and overall life satisfaction in data from Canada’s General Social Survey for the 2005 and 2010 cycles. We find that here, consistent with the evidence in a number of other studies, “money isn’t everything” – the direct effect of income on life satisfaction is modest. Considering the possibility of an indirect effect of income through financial stress, we find evidence of a strong effect of financial stress on life satisfaction. Considering the certainty with which high financial stress

reduces the likelihood of reporting high financial satisfaction suggests that for financially stressed Canadians it is the financial domain which dominates to a considerable degree one's assessment of life satisfaction.

We find financial stress and financial dissatisfaction to be closely related. Specifically, we find that high financially stressed households are not reporting the highest financial satisfaction with certainty. While financial stress is not the same thing as financial satisfaction in the results we observe, the certainty of the connection between high financial stress and lower financial satisfaction together with the considerable effect of financial stress on life satisfaction (and considerably higher than any other determinant) supports the suggestion that the domain of financial satisfaction overwhelmingly influences life satisfaction in financially stressed households, swamping even health and employment determinants.

Allowing for the possibility that financial stress may play more than a mediating role, we explored as well the possibility of financial stress conditioning or moderating the relationship for individuals between their income and their life satisfaction. That is, we explored whether financial stress differentially influences the effect of income on life satisfaction. Here, we find only a limited moderating role for financial stress. The more financially stressed the high income earners, the lower the probability that the higher income will enhance their sense of life satisfaction. In 2005 (but not in 2010) accounting for the interaction between income and financial stress reveals the modest but statistically significant moderating effect of high income on life satisfaction.

The implication of these results for policy is that the considerable influence of subjective financial stress in affecting overall life satisfaction begs the question about its relationship to the principal foci of policy makers – objective financial stress and financial literacy. How does subjective financial stress relate to indicators of objective financial stress such as debt loads and financial behaviour flowing from levels of literacy? Since policy makers have been most concerned with objective financial stress indicators and more recently increased their focus on financial behaviour and levels of financial literacy, the relationship between subjective stress and objective stress as either mediated or moderated or both by levels of financial literacy is a question we leave to further research.

In short, while the evidence examined in this study cannot support a strict interpretation of Kanye West's subordinate phrase, we find important evidence to suggest that financial stress as the *perception* of "not havin' money" is, indeed, (almost) everything. Understanding how best to confront this perception of inadequacy may improve policy effectiveness and will enhance life satisfaction considerably.

Table 1. Multinomial probits for financial and non-financial stress		
STRESS	Financial stress 2005 & 2010 GSS	Non-financial stress 2005 & 2010 GSS
Specification	(1-FS)	(1-NFS)
α	2005 Nominal household income less than 20K	0.347 (0.199)
	2005 Nominal household income greater than 100K	-0.330** (0.085)
	2010 Nominal household income less than 20K	0.077 (0.065)
	2010 Nominal household income greater than 100K	-0.406** (0.032)
γ	High school dropout	-0.002 (0.025)
	University of higher	-0.408** (0.045)
	Age 15 – 24	-0.065 (0.064)
	Age > 64	-0.428** (0.058)
	Respondent is an income provider	0.014 (0.040)
	Part – time employed (less than 30 hours per week)	0.424** (0.074)
	Student (employed or not employed)	-0.619** (0.109)
	Not employed	0.364** (0.070)
	Female	-0.236** (0.028)
	Widowed/separated/divorced	0.318** (0.054)
	Single (never married)	0.137* (0.061)
	Children under 19 present	0.146* (0.071)
	Household size	0.013 (0.022)
	Constant	-1.252** (0.279)

1. *5%, **1%
2. Results conditional on a full set of provincial dummies, urban area of residence dummy, multigenerational household and on interview conducted on weekend.
3. Default categories: No stress, high school and college education, age 35 – 44, respondent not an income provider, full time employed, male, married, no children under 19 living, in the house, not a multigenerational household, one household member, Quebec, rural area of residence, nominal household income between 20K and 100K, weekday.
4. Bolded income coefficients are used in the formal test of moderation.

Table 2. Ordered probits for LIFE Satisfaction and FINANCIAL Satisfaction

SATISFACTION	Ordered probit for LIFE satisfaction Sample (2005 and 2010 GSS)			Ordered probit for FINANCIAL satisfaction Sample (2005 GSS)		
	All respondents	All respondents	All respondents	All respondents	All respondents	All respondents
Specification	2-L	3-L	4-L	2-F	3-F	4-F
2005 Nominal household income less than 20K	-0.234** (0.039)	-0.205** (0.024)	-0.238** (0.041)	-0.455** (0.090)	-0.431** (0.070)	-0.453** (0.022)
2005 Nominal household income greater than 100K	0.119** (0.022)	0.101** (0.019)	-0.0002 (0.029)	0.264** (0.038)	0.239** (0.030)	0.147** (0.038)
2010 Nominal household income less than 20K	-0.143** (0.041)	-0.116* (0.048)	0.092 (0.094)			
2010 Nominal household income greater than 100K	0.153** (0.020)	0.153** (0.017)	0.173** (0.042)			
β Financial stress		-0.864** (0.044)	-0.900** (0.062)		-1.030** (0.053)	-1.090** (0.037)
Non-Financial stress		-0.563** (0.029)	-0.580** (0.033)		-0.256** (0.025)	-0.322** (0.020)
(Financial stress) x (2005 Nominal household income less than 20K)			0.134 (0.134)			0.053 (0.044)
(Financial stress) x (2005 Nominal household income greater than 100K)			0.239** (0.082)			0.138 (0.088)
δ (Non-Financial stress) x (2005 Nominal household income less than 20K)			0.030 (0.096)			0.018 (0.123)
(Non-Financial stress) x (2005 Nominal household income greater than 100K)			0.162** (0.036)			0.157** (0.026)
(Financial stress) x (2010 Nominal household income less than 20K)			-0.231 (0.145)			
(Financial stress) x (2010 Nominal household income greater than 100K)			-0.022 (0.108)			
(Non-Financial stress) x (2010 Nominal household income less than 20K)			-0.412** (0.102)			
(Non-Financial stress) x (2010 Nominal household income greater than 100K)			-0.035 (0.064)			

Table 2. continued

SATISFACTION	Ordered probit for LIFE satisfaction Sample (2005 and 2010 GSS)			Ordered probit for FINANCIAL satisfaction Sample (2005 GSS)		
	All respondents	All respondents	All respondents	All respondents	All respondents	All respondents
Specification	2-L	3-L	4-L	2-F	3-F	4-F
High school dropout	-0.011 (0.022)	-0.021 (0.021)	-0.023 (0.021)	-0.045* (0.020)	-0.048* (0.020)	-0.049* (0.020)
University of higher	0.005 (0.015)	0.028 (0.015)	0.022 (0.015)	0.127** (0.016)	0.118** (0.013)	0.116** (0.013)
Age 15 – 24	0.239** (0.034)	0.201** (0.041)	0.205** (0.041)	0.100** (0.023)	0.068** (0.020)	0.072** (0.020)
Age > 64	0.420** (0.042)	0.304** (0.036)	0.303** (0.035)	0.441** (0.022)	0.356** (0.020)	0.358** (0.020)
Respondent is an income provider	0.021 (0.021)	0.048* (0.022)	0.043* (0.021)	-0.038 (0.034)	-0.022 (0.033)	-0.026** (0.032)
Part-time employed (less than 30 hours per week)	0.074** (0.018)	0.003 (0.024)	0.002 (0.023)	-0.027 (0.036)	-0.047 (0.035)	-0.049 (0.034)
Student (employed or not employed)	0.091* (0.051)	0.067 (0.044)	0.070 (0.042)	-0.130** (0.040)	-0.203** (0.044)	-0.203** (0.045)
Not employed	-0.033 (0.028)	-0.154 (0.024)	-0.154** (0.022)	-0.059* (0.025)	-0.122** (0.021)	-0.123** (0.022)
Female	0.031* (0.014)	0.081** (0.016)	0.083** (0.016)	0.009 (0.017)	0.031 (0.019)	0.033 (0.019)
Widowed/separated/divorced	-0.364** (0.020)	-0.359** (0.015)	-0.361** (0.015)	-0.381** (0.012)	-0.351** (0.016)	-0.348** (0.016)
Single (never married)	-0.302** (0.016)	-0.299** (0.021)	-0.301** (0.021)	-0.340** (0.021)	-0.319** (0.019)	-0.319** (0.019)
Children under 19 present	-0.047** (0.018)	0.007 (0.020)	0.008 (0.020)	-0.091** (0.023)	-0.051* (0.024)	-0.050* (0.023)
Household size	-0.008 (0.011)	-0.005 (0.011)	-0.006 (0.011)	-0.029** (0.008)	-0.025** (0.007)	-0.025** (0.007)

1. *5%, **1%

2. Results conditional on full set of provincial dummies, urban area of residence dummy, multigenerational household and weekend interview.

3. Default categories: No stress, high school and college education, age 35 – 44, respondent not an income provider, full time employed, male, married, no children under 19 living, in the house, not a multigenerational household, one household member, Quebec, rural area of residence, nominal household income between 20K and 100K, weekday.

4. Bolded financial stress coefficients are used in the formal test of moderation.

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REFERENCES

- Ayllón, S., & Fusco, A. (2017). Are income poverty and perceptions of financial difficulties dynamically interrelated?. *Journal of Economic Psychology*, 61, 103-114.
- Bailey, A. W. (1987). Social and economic factors affecting the financial well-being of families. *Journal of Home Economics*, 79(2), 14-18.
- Bailey, W. C., Woodiel, D. K., Turner, M. J., & Young, J. (1998). The relationship of financial stress to overall stress and satisfaction. *Personal Finances and Worker Productivity*, 2(2), 198-207.
- Barrington-Leigh, C. P., & Helliwell, J. F. (2008). Empathy and emulation: Life satisfaction and the urban geography of comparison groups (No. w14593). National Bureau of Economic Research. On line at <http://www.nber.org/papers/w14593>
- Bonikowska, A., Helliwell, J. F., Hou, F., & Schellenberg, G. (2014). An assessment of life satisfaction responses on recent Statistics Canada surveys. *Social Indicators Research*, 118(2), 617-643.
- Brown, S., & Gray, D. (2016). Household finances and well-being in Australia: An empirical analysis of comparison effects. *Journal of Economic Psychology*, 53, 17-36.
- Christoph, B. (2010). The relation between life satisfaction and the material situation: A re-evaluation using alternative measures. *Social Indicators Research*, 98(3), 475-499.
- D'Ambrosio, C., & Frick, J. R. (2007). Income satisfaction and relative deprivation: An empirical link. *Social Indicators Research*, 81(3), 497-519.
- Danes, S. M., & Rettig, K. D. (1993). The role of perception in the intention to change the family financial situation. *Journal of Family and Economic Issues*, 14(4), 365-389.
- Diener, E. 1984. Subjective well-being. *Psychological Bulletin* 95(3), 542-575.
- Diener, E., Sandvik, E., Seidlitz, L., & Diener, M. (1993). The relationship between income and subjective well-being: Relative or absolute?. *Social Indicators Research*, 28(3), 195-223
- Diener, E., Suh, E. M., Lucas, R. E., & Smith, H. (1999). Subjective well-being: Three decades of progress. *Psychological Bulletin*, 125, 276-302.
- Diener E, Ng W, Harter J, & Arora R (2010) Wealth and happiness across the world: Material prosperity predicts life evaluation, whereas psychosocial prosperity predicts positive feeling. *Journal of Personality and Social Psychology*, 99(1), 52-61.
- Dolan, P., Peasgood, T., & White, M. (2008). Do we really know what makes us happy? A review of the economic literature on the factors associated with subjective well-being. *Journal of Economic Psychology*, 29(1), 94-122.
- Easterlin, R. A. (2006). Life cycle happiness and its sources: Intersections of psychology, economics, and demography. *Journal of Economic Psychology*, 27(4), 463-482.
- Easterlin, R. A. (2015). Happiness and economic growth—the evidence. In *Global handbook of quality of life* (pp. 283-299). Springer Netherlands.
- Fusco, A. (2016) The Dynamics of Perceived Financial Difficulties. *Journal of Happiness Studies*, 17, 1599-1614 DOI 10.1007/s10902-015-9661-5
- Garðarsdóttir, R. B., & Dittmar, H. (2012). The relationship of materialism to debt and financial well-being: The case of Iceland's perceived prosperity. *Journal of Economic Psychology*, 33(3), 471-481.
- Gasiorowska, A. (2014). The relationship between objective and subjective wealth is moderated by financial control and mediated by money anxiety. *Journal of Economic Psychology*, 43, 64-74.
- Grable, J. E., Cupples, S., Fernatt, F., & Anderson, N. (2013). Evaluating the link between perceived income adequacy and financial satisfaction: A resource deficit hypothesis approach. *Social Indicators Research*, 114(3), 1109-1124.
- Hayes, A. F. (2012). PROCESS: A versatile computational tool for observed variable mediation, moderation, and conditional process modeling [White paper]. Retrieved from <http://www.afhayes.com/public/process2012.pdf>
- Hayhoe, C. R., Cho, S. H., DeVaney, S. A., Worthy, S. L., Kim, J., & Gorham, E. (2012). How do distrust and anxiety affect saving behavior?. *Family and Consumer Sciences Research Journal*, 41(1), 69-85.
- Heller, D., Watson, D., & Ilies, R. (2004). The role of person versus situation in life satisfaction: a critical examination. *Psychological Bulletin*, 130(4), 574-600
- Hilgert, M. A., Hogarth, J. M., & Beverly, S. G. (2003). Household financial management: The connection between knowledge and behavior. *Federal Reserve Bulletin*, 89, 309-322.

- Hira, T. K., & Loibl, C. (2005). Understanding the impact of employer-provided financial education on workplace satisfaction. *Journal of Consumer Affairs*, 39(1), 173-194.
- Hsieh, C. M. (2004). Income and financial satisfaction among older adults in the United States. *Social Indicators Research*, 66(3), 249-266.
- Johnson, W., & Krueger, R. F. (2006). How money buys happiness: genetic and environmental processes linking finances and life satisfaction. *Journal of Personality and Social Psychology*, 90(4), 680-691.
- Joo, S. H., & Grable, J. E. (2004). An exploratory framework of the determinants of financial satisfaction. *Journal of family and economic Issues*, 25(1), 25-50.
- Keese, M. (2012). Who feels constrained by high debt burdens? Subjective vs. objective measures of household debt. *Journal of Economic Psychology*, 33(1), 125-141.
- Latif, E. (2016). Happiness and comparison income: evidence from Canada. *Social Indicators Research*, 128(1), 161-177. Doi: 10.1007/s11205-015-1024-4
- Lusardi, A. (2008). Household saving behavior: The role of financial literacy, information, and financial education programs (No. w13824). National Bureau of Economic Research.
- Mullainathan, S., & Shafir, E. (2013). *Scarcity: Why having too little means so much*. Macmillan.
- Ng, W., & Diener, E. (2014). What matters to the rich and the poor? Subjective well-being, financial satisfaction, and postmaterialist needs across the world. *Journal of Personality and Social Psychology*, 107(2), 326-338.
- O'Neill, B., Sorhaindo, B., Xiao, J. J., & Garman, E. T. (2005). Financially distressed consumers: Their financial practices, financial well-being, and health. *Journal of Financial Counseling and Planning*, 16(1).
- Pereira, M. C., & Coelho, F. (2013). Untangling the relationship between income and subjective well-being: The role of perceived income adequacy and borrowing constraints. *Journal of Happiness Studies*, 14(3), 985-1005.
- Plagnol, A. (2011). Financial satisfaction over the life course: The influence of assets and liabilities. *Journal of Economic Psychology*, 32(1), pp. 45-64.
- Porter, N.M., & Garman, E.T. (1993). Testing a conceptual model of financial well-being. *Financial Counseling and Planning*, 4, 135-164
- Prawitz, A. D., Garman, E. T., Sorhaindo, B., O'Neill, B., Kim, J., & Drentea, P. (2006). In Charge financial distress/financial well-being scale: Development, administration, and score interpretation. *Financial Counseling and Planning* 17(1) 34-50.
- Sahi, S. K. (2017). Psychological biases of individual investors and financial satisfaction. *Journal of Consumer Behaviour*. <https://doi.org/10.1002/cb.1644>
- Schwarz, N., Strack, F., & Mai, H. P. (1991). Assimilation and contrast effects in part-whole question sequences: A conversational logic analysis. *Public Opinion Quarterly*, 55, 3-23.
- Sinclair, R., Sears, L. E., Probst, T., & Zajack, M. (2010). A multilevel model of economic stress and employee well-being. In J. Houdmont, J., & S. Leka, (Eds.) *Contemporary Occupational Health Psychology: Global Perspectives on Research and Practice Volume 1* (pp. 1-20). John Wiley & Sons
- Statistics Canada (2013) An Assessment of Life Satisfaction Responses on Recent Statistics Canada Surveys, Available at <http://www.statcan.gc.ca/pub/11f0019m/2013351/part-partie1-eng.htm>
- Van Praag, B. M., Frijters, P., & Ferrer-i-Carbonell, A. (2003). The anatomy of subjective well-being. *Journal of Economic Behavior & Organization*, 51(1), 29-49.